```
akankshya@LAPTOP-84O9I4VO:~$ tree ~
/home/akankshya
├── akankshya.txt
├── linux-practice
│   ├── akankshya.txt
│   ├── my
│   │   └── folder
│   │       └── one
│   ├── notes.txt
│   └── renamed.txt
├── my
│   └── folder
│       ├── 1.txt
│       ├── akankshya.txt
│       └── one
├── my-first-website
├── my-folder
├── oss-project
│   ├── manifesto.txt
│   ├── manifesto_akankshya.txt
│   ├── script1.sh
│   ├── script2.sh
│   ├── script3.sh
│   ├── script4.sh
│   └── script5.sh
└── script1.sh.save

11 directories, 14 files
```

```
akankshya@LAPTOP-84O9I4V
akankshya@LAPTOP-84O9I4VO:/mnt/c/Users/akank$ cd /
akankshya@LAPTOP-84O9I4VO:/$ pwd
/
akankshya@LAPTOP-84O9I4VO:/$ cd /home
akankshya@LAPTOP-84O9I4VO:/home$ ls
akankshya
akankshya@LAPTOP-84O9I4VO:/home$ /home/akankshya
-bash: /home/akankshya: Is a directory
akankshya@LAPTOP-84O9I4VO:/home$ /root
-bash: /root: Is a directory
akankshya@LAPTOP-84O9I4VO:/home$ ~
-bash: /home/akankshya: Is a directory
akankshya@LAPTOP-84O9I4VO:/home$ pwd
/home
akankshya@LAPTOP-84O9I4VO:/home$ ls
akankshya
akankshya@LAPTOP-84O9I4VO:/home$ ls -l
total 4
drwxr-x--- 7 akankshya akankshya 4096 Nov  3 10:06 akankshya
akankshya@LAPTOP-84O9I4VO:/home$ ls -a
.  ..  akankshya
akankshya@LAPTOP-84O9I4VO:/home$ ls -la
total 12
drwxr-xr-x  3 root      root      4096 Mar 22  2026 .
drwxr-xr-x 27 root      root      4096 Nov  3 14:01 ..
drwxr-x---  7 akankshya akankshya 4096 Nov  3 10:06 akankshya
akankshya@LAPTOP-84O9I4VO:/home$ cd
akankshya@LAPTOP-84O9I4VO:~$ cd /home
akankshya@LAPTOP-84O9I4VO:/home$ cd ~
akankshya@LAPTOP-84O9I4VO:~$ pwd
/home/akankshya
akankshya@LAPTOP-84O9I4VO:~$ cd ..
akankshya@LAPTOP-84O9I4VO:/home$ pwd
/home
akankshya@LAPTOP-84O9I4VO:/home$ cd -
/home/akankshya
akankshya@LAPTOP-84O9I4VO:~$ pwd
/home/akankshya
akankshya@LAPTOP-84O9I4VO:~$ cd /
akankshya@LAPTOP-84O9I4VO:/$ pwd
/
akankshya@LAPTOP-84O9I4VO:/$ history
    1  cd ~
    2  mkdir oss-project
    3  cd oss-project
    4  wsl
    5  touch script1.sh script2.sh script3.sh script4.sh script5.sh
    6  nano script1.sh
    7  nano script2.sh
    8  nano script1.sh
    9  nano script2.sh
   10  chmod +x script2.sh
   11  ./script2.sh
   12  nano script3.sh
   13  chmod +x script3.sh
   14  ./script3.sh
   15  nano script4.sh
   16  chmod +x script4.sh
   17  ./script4.sh
   18  nano script5.sh
   19  chmod +x script5.sh
```

```
  233  ~
  234  pwd
  235  ls
  236  ls -l
  237  ls -a
  238  ls -la
  239  cd
  240  cd /home
  241  cd ~
  242  pwd
  243  cd ..
  244  pwd
  245  cd -
  246  pwd
  247  cd /
  248  pwd
  249  history
akankshya@LAPTOP-84O9I4VO:/$
```

```
akankshya@LAPTOP-84O9I4VO:/$ pwd
/
akankshya@LAPTOP-84O9I4VO:/$ cd ~
akankshya@LAPTOP-84O9I4VO:~$ pwd
/home/akankshya
akankshya@LAPTOP-84O9I4VO:~$ mkdir my-folder
akankshya@LAPTOP-84O9I4VO:~$ pwd
/home/akankshya
akankshya@LAPTOP-84O9I4VO:~$ ls
akankshya.txt  my  my-first-website  my-folder  oss-project  script1.sh.save
akankshya@LAPTOP-84O9I4VO:~$ mkdir linux-practice
akankshya@LAPTOP-84O9I4VO:~$ ls
akankshya.txt   my                my-folder    script1.sh.save
linux-practice  my-first-website  oss-project
akankshya@LAPTOP-84O9I4VO:~$ cd linux-practice
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ pwd
/home/akankshya/linux-practice
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ mkdir -p my/folder/one
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ pwd
/home/akankshya/linux-practice
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ ls
my
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ mkdir -p my/folder/one
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ ls
my
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ pwd
/home/akankshya/linux-practice
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ ls
my
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ touch 1.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ ls
1.txt  my
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ echo "Hello Akankshya" > akankshya.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ ls
1.txt  akankshya.txt  my
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ echo "I am learning Linux" >> akankshya.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ ls
1.txt  akankshya.txt  my
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cat > notes.txt
Linux is an operating system.
I am learning Linux commands.
I am practising WSL Ubuntu.akankshya@LAPTOP-84O9I4VO:~/linux-practice$
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cat notes.txt
Linux is an operating system.
I am learning Linux commands.
I am practising WSL Ubuntu.akankshya@LAPTOP-84O9I4VO:~/linux-practice$ vim notes.txt
```

```
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cat akankshya.txt
Hello Akankshya
I am learning Linux
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cat -n akankshya.txt
     1  Hello Akankshya
     2  I am learning Linux
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ less akankshya.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ head akankshya.txt
Hello Akankshya
I am learning Linux
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ tail akankshya.txt
Hello Akankshya
I am learning Linux
```

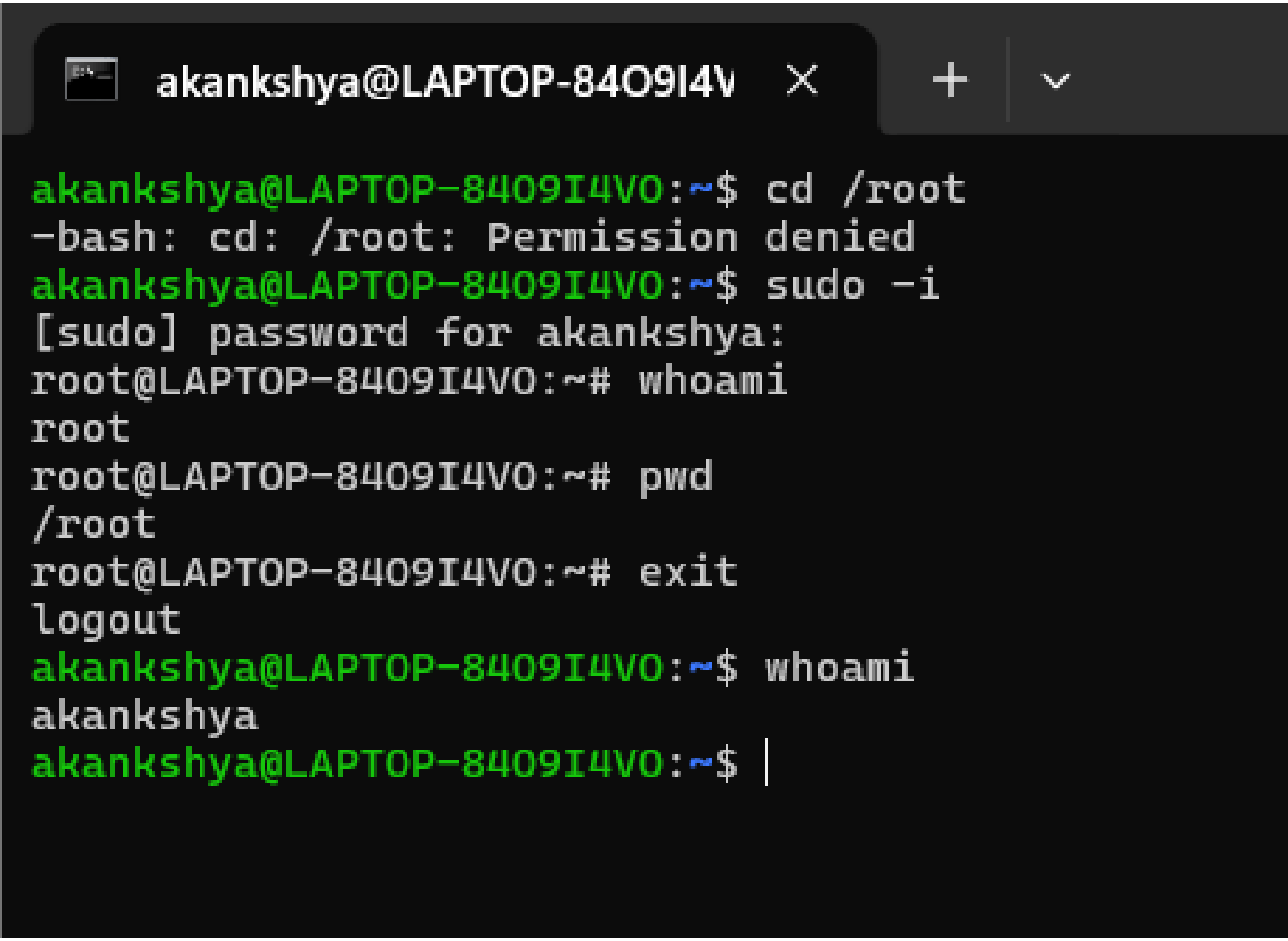
```
akankshya@LAPTOP-84O9I4VO:~/linux-practice/my$ ls
folder
akankshya@LAPTOP-84O9I4VO:~/linux-practice/my$ cd ..
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ pwd
/home/akankshya/linux-practice
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ ls
1.txt  akankshya.txt  my  notes.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ ls
1.txt  akankshya.txt  my  notes.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cp 1.txt akankshya.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cat akankshya.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ echo "Hello Akankshya" > 1.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cat 1.txt
Hello Akankshya
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cp 1.txt akankshya.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cat akankshya.txt
Hello Akankshya
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cat 1.txt > akankshya.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ cat akankshya.txt
Hello Akankshya
akankshya@LAPTOP-84O9I4VO:~/linux-practice$
```

```
akankshya@LAPTOP-84O9I4V
akankshya@LAPTOP-84O9I4VO:~$ cd ~
akankshya@LAPTOP-84O9I4VO:~$ find ~/linux-practice -name "1.txt"
akankshya@LAPTOP-84O9I4VO:~$ ls -R ~/linux-practice
/home/akankshya/linux-practice:
akankshya.txt  my  newfile.txt  notes.txt

/home/akankshya/linux-practice/my:
folder

/home/akankshya/linux-practice/my/folder:
one

/home/akankshya/linux-practice/my/folder/one:
akankshya@LAPTOP-84O9I4VO:~$ find ~/linux-practice -type f -name "*.txt"
/home/akankshya/linux-practice/notes.txt
/home/akankshya/linux-practice/akankshya.txt
/home/akankshya/linux-practice/newfile.txt
akankshya@LAPTOP-84O9I4VO:~$ pwd
/home/akankshya
akankshya@LAPTOP-84O9I4VO:~$ cd ~/linux-practice
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ mv newfile.txt renamed.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$ ls
akankshya.txt  my  notes.txt  renamed.txt
akankshya@LAPTOP-84O9I4VO:~/linux-practice$
```


akankshya@LAPTOP-84O9I4V
akankshya@LAPTOP-84O9I4VO:~$ cd /root
-bash: cd: /root: Permission denied
akankshya@LAPTOP-84O9I4VO:~$ sudo -i
[sudo] password for akankshya:
root@LAPTOP-84O9I4VO:~# whoami
root
root@LAPTOP-84O9I4VO:~# pwd
/root
root@LAPTOP-84O9I4VO:~# exit
logout
akankshya@LAPTOP-84O9I4VO:~$ whoami
akankshya
akankshya@LAPTOP-84O9I4VO:~$

Command	Purpose	Example
`pwd`	Show current location	`pwd`
`ls`	List files and folders	`ls`
`ls -l`	Detailed listing	`ls -l`
`ls -a`	Show hidden files	`ls -a`
`cd`	Change directory	`cd my`
`cd ..`	Go one level up	`cd ..`
`cd ~`	Go home	`cd ~`
`cd /`	Go to top-level directory	`cd /`
`clear`	Clear terminal	`clear`
`history`	Show previous commands	`history`
`mkdir`	Create directory	`mkdir test`
`mkdir -p`	Create nested directories	`mkdir -p a/b/c`
`touch`	Create empty file	`touch a.txt`
`echo`	Print text	`echo "Hello"`

`cat`	Display file content	`cat a.txt`
`cat -n`	Display with line numbers	`cat -n a.txt`
`cp`	Copy files	`cp a.txt b.txt`
`mv`	Move or rename files	`mv a.txt b.txt`
`rm`	Delete a file	`rm a.txt`
`rmdir`	Delete empty directory	`rmdir test`
`vim`	Open text editor	`vim a.txt`
`less`	Read file page by page	`less a.txt`
`head`	Display first lines	`head a.txt`
`tail`	Display last lines	`tail a.txt`
`whoami`	Show current user	`whoami`
`sudo`	Run command with elevated privileges	`sudo apt update`
`sudo -i`	Enter root shell	`sudo -i`
`exit`	Exit root shell or current shell	`exit`
`chmod`	Change permissions	`chmod +x file.sh`

```
akankshya@LAPTOP-84O9I4VO:~$ sudo -i
[sudo] password for akankshya:
root@LAPTOP-84O9I4VO:~# whoami
root
root@LAPTOP-84O9I4VO:~# sudo adduser gullu
info: Adding user `gullu' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `gullu' (1001) ...
info: Adding new user `gullu' (1001) with group `gullu (1001)' ...
info: Creating home directory `/home/gullu' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for gullu
Enter the new value, or press ENTER for the default
        Full Name []: Aka Rout
        Room Number []: 1
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `gullu' to supplemental / extra groups `users' ...
info: Adding user `gullu' to group `users' ...
root@LAPTOP-84O9I4VO:~# su - gullu
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Nov  3 17:02:23 UTC 2026

  System load:  0.0                 Processes:             37
  Usage of /:   0.2% of 1006.85GB   Users logged in:       2
  Memory usage: 5%                  IPv4 address for eth0: 172.18.137.11
  Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

This message is shown once a day. To disable it please create the
/home/gullu/.hushlogin file.
gullu@LAPTOP-84O9I4VO:~$ sudo apt update
[sudo] password for gullu:
gullu is not in the sudoers file.
gullu@LAPTOP-84O9I4VO:~$ whoami
gullu
gullu@LAPTOP-84O9I4VO:~$ pwd
/home/gullu
gullu@LAPTOP-84O9I4VO:~$ ls
```

```
gullu@LAPTOP-84O9I4VO:~$ sudo -i
[sudo] password for gullu:
gullu is not in the sudoers file.
gullu@LAPTOP-84O9I4VO:~$ whoami
gullu
gullu@LAPTOP-84O9I4VO:~$ pwd
/home/gullu
gullu@LAPTOP-84O9I4VO:~$ ls
gullu@LAPTOP-84O9I4VO:~$ exit
logout
root@LAPTOP-84O9I4VO:~# sudo usermod -aG sudo gullu
root@LAPTOP-84O9I4VO:~# su - gullu
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

gullu@LAPTOP-84O9I4VO:~$ sudo apt update
[sudo] password for gullu:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1155 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [233 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1224 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [247 kB]
Fetched 2984 kB in 5s (609 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
206 packages can be upgraded. Run 'apt list --upgradable' to see them.
gullu@LAPTOP-84O9I4VO:~$
```

```
akankshya@LAPTOP-84O9I4VO:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
206 packages can be upgraded. Run 'apt list --upgradable' to see them.
akankshya@LAPTOP-84O9I4VO:~$ sudo apt install apache2
```

```
akankshya@LAPTOP-84O9I4VO:~$ sudo apt remove apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap
  libaprutil1t64 liblua5.4-0 ssl-cert
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  apache2
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 468 kB disk space will be freed.
Do you want to continue? [Y/n] y
(Reading database ... 41517 files and directories currently installed.)
Removing apache2 (2.4.58-1ubuntu8.17) ...
Processing triggers for man-db (2.12.0-4build2) ...
akankshya@LAPTOP-84O9I4VO:~$
```

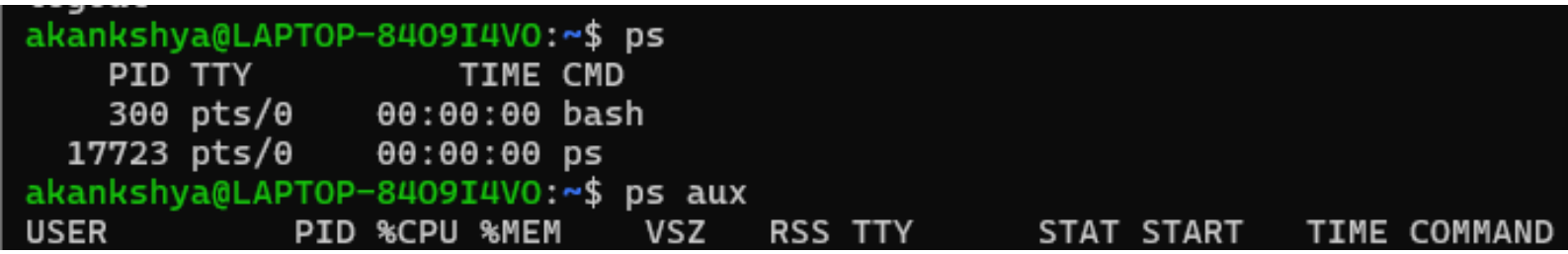

akankshya@LAPTOP-8409I4VO:~$ sudo apt show nginx

```
akankshya@LAPTOP-84O9I4VO:~$ sudo apt install curl git python3
```

```
akankshya@LAPTOP-84O9I4VO:~$ apt list -- install
Listing... Done
akankshya@LAPTOP-84O9I4VO:~$ sudo apt-get update
```

```
akankshya@LAPTOP-84O9I4VO:~$ whoami
akankshya
akankshya@LAPTOP-84O9I4VO:~$ sudo -i
root@LAPTOP-84O9I4VO:~# whoami
root
root@LAPTOP-84O9I4VO:~# sudo groupadd developers
root@LAPTOP-84O9I4VO:~# sudo useradd -m rohan
root@LAPTOP-84O9I4VO:~# sudo useradd -m lovish
root@LAPTOP-84O9I4VO:~# sudo passwd rohan
New password:
Retype new password:
passwd: password updated successfully
root@LAPTOP-84O9I4VO:~# sudo passwd lovish
New password:
Retype new password:
passwd: password updated successfully
root@LAPTOP-84O9I4VO:~# cd /home
root@LAPTOP-84O9I4VO:/home# ls
akankshya  gullu  lovish  rohan
root@LAPTOP-84O9I4VO:/home# sudo usermod -aG developers rohan
root@LAPTOP-84O9I4VO:/home# sudo usermod -aG developers lovish
root@LAPTOP-84O9I4VO:/home# groups rohan
rohan : rohan developers
root@LAPTOP-84O9I4VO:/home# groups lovish
lovish : lovish developers
root@LAPTOP-84O9I4VO:/home# ls -l
total 16
drwxr-x--- 9 akankshya akankshya 4096 Nov  3 16:08 akankshya
drwxr-x--- 3 gullu     gullu     4096 Nov  3 17:11 gullu
drwxr-x--- 2 lovish    lovish    4096 Nov  3 17:45 lovish
drwxr-x--- 2 rohan     rohan     4096 Nov  3 17:45 rohan
root@LAPTOP-84O9I4VO:/home# cd gullu
root@LAPTOP-84O9I4VO:/home/gullu# mkdir devs
root@LAPTOP-84O9I4VO:/home/gullu# ls -l
total 4
drwxr-xr-x 2 root root 4096 Nov  3 17:50 devs
root@LAPTOP-84O9I4VO:/home/gullu# sudo chown lovish devs
root@LAPTOP-84O9I4VO:/home/gullu# ls -l
total 4
drwxr-xr-x 2 lovish root 4096 Nov  3 17:50 devs
root@LAPTOP-84O9I4VO:/home/gullu# sudo chgrp developers devs
root@LAPTOP-84O9I4VO:/home/gullu# ls -l
total 4
drwxr-xr-x 2 lovish developers 4096 Nov  3 17:50 devs
root@LAPTOP-84O9I4VO:/home/gullu#
root@LAPTOP-84O9I4VO:/home/gullu# chmod g+w devs
root@LAPTOP-84O9I4VO:/home/gullu# ls -l
total 4
drwxrwxr-x 2 lovish developers 4096 Nov  3 17:50 devs
root@LAPTOP-84O9I4VO:/home/gullu# chmod o+w devs
root@LAPTOP-84O9I4VO:/home/gullu# ls -l
total 4
drwxrwxrwx 2 lovish developers 4096 Nov  3 17:50 devs
root@LAPTOP-84O9I4VO:/home/gullu#
```

```
root@LAPTOP-84O9I4VO:/home/gullu# chmod 435 devs
root@LAPTOP-84O9I4VO:/home/gullu# ls -l
total 4
dr---wxr-x 2 lovish developers 4096 Nov  3 17:50 devs
root@LAPTOP-84O9I4VO:/home/gullu#
```


akankshya@LAPTOP-84O9I4VO:~$ ps
PID TTY TIME CMD
300 pts/0 00:00:00 bash
17723 pts/0 00:00:00 ps
akankshya@LAPTOP-84O9I4VO:~$ ps aux
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND